

Mohamad Khalif

Portfolio

GEODETIC ENGINEER

Introduction

I'm a Geodetic Engineering fresh graduate, a litle mix between a researcher and a field practitioner. I'mquite adaptive, can work independently but also enjoy teamwork. I love problem-solving, critical thinking, andpaying attention to details is really important for me. I'm also the type who's open to feedback and learns newthings quickly whether it's tools, workflow, or project challenges. I have experience in several academic andfield projects. 'm very interested in terrestrial surveying, remote sensing, and cloud-based data platforms likeGoogle Earth Engine to create interactive web-based maps. Besides that, 'm also interested in A/ML, especially ingeospatial analysis. During college, P've been involved in various projects such as terrestrial surveys, hydrography,aerial mapping, and of course remote sensing as my favorite field. During my internship in the Probowangi Toll Roadproject, I handed staking out points and land volume calculations for cut-and-fill analysis based on field

coordinates.Throughout my academic journey, I'm quite fluent in using geospatial software such as ArcMap, QGIS, andENV. Coding is also an important part of my work: Python, PGAdmin, Java (Google Earth Engine), and MATLAB aremy daily tools. I have used TensorFlow & Keras to build vegetation classification models from high-resolutionsatelite imagery, and in my thesis, I successfully analyzed and created a mulitemporal aerosol distribution map ofJakarta (2021-2024) using Sentinel and MATLAB. My friends usually call me "the vibe coder" because I loveexploring new things and finding ways to blend geospatial science, remote sensing, and AV/ML into impactfulsolutions.



PT. Virama Karya (Probowangi Toll Road Package 1)

During this internship, I served as a surveyor assistant, responsible for staking out the Top Subgrade, Lean Concrete, and Rigid Pavement to ensure precise alignment with the planned coordinates and elevations.
In my final internship report, I performed land volume calculations using long and cross-section maps to identify cut and fill areas based on the marking coordinates.



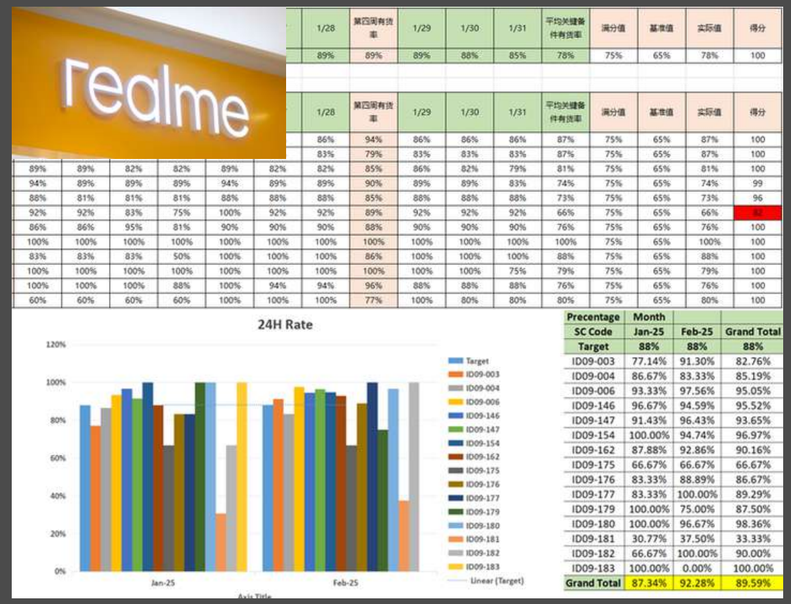
PTSL (Pendaftaran Tanah Sistematis Lengkap) Lamongan, Kec. Solokuro

In this experience, I was given the opportunity to participate in the PTSL program with a project to map several village boundaries with local residents and determine and supervise the installation of ICP (Independent Control Point) markers for photography from a UAV.



PT. PT Thousand Island Services

I am experienced in handing end-to-end warehouse administration processes including stock opname, returnmanagement, claim processing, work order monitoring, and unit replacement coordination. I also manage inter-agent spare part transfers to optimize stock distribution and minimize shortages.



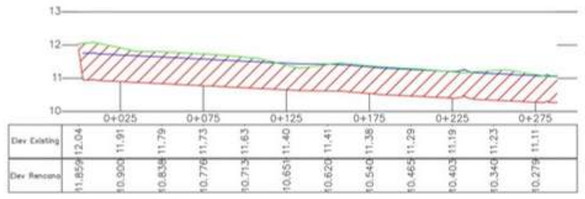
TELUS Digital International

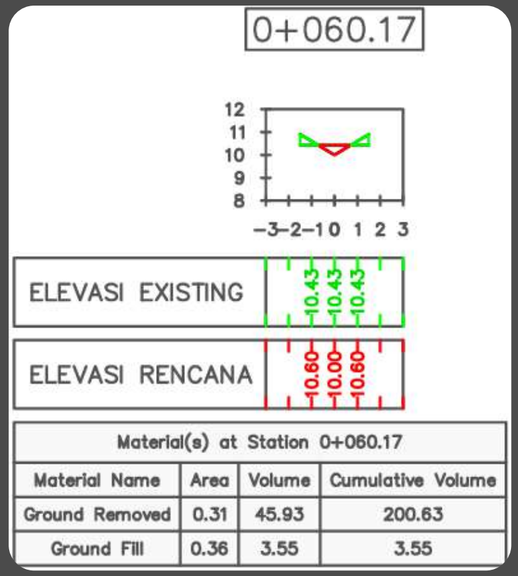
I have experience working at TELUS International in digital content evaluation and analysis, where my responsibilities included reviewing and evaluating information based on established guidelines and quality criteria. This role strengthened my ability to analyze information systematically, maintain accuracy and consistency in evaluation results, and follow standard operating procedures. I am also accustomed to working with performance targets, learning new guidelines, and maintaining high-quality work standards. This experience has developed my analytical skills, attention to detail, and discipline, which I can apply to spatial data processing, operational map updates, and other GIS-related tasks.



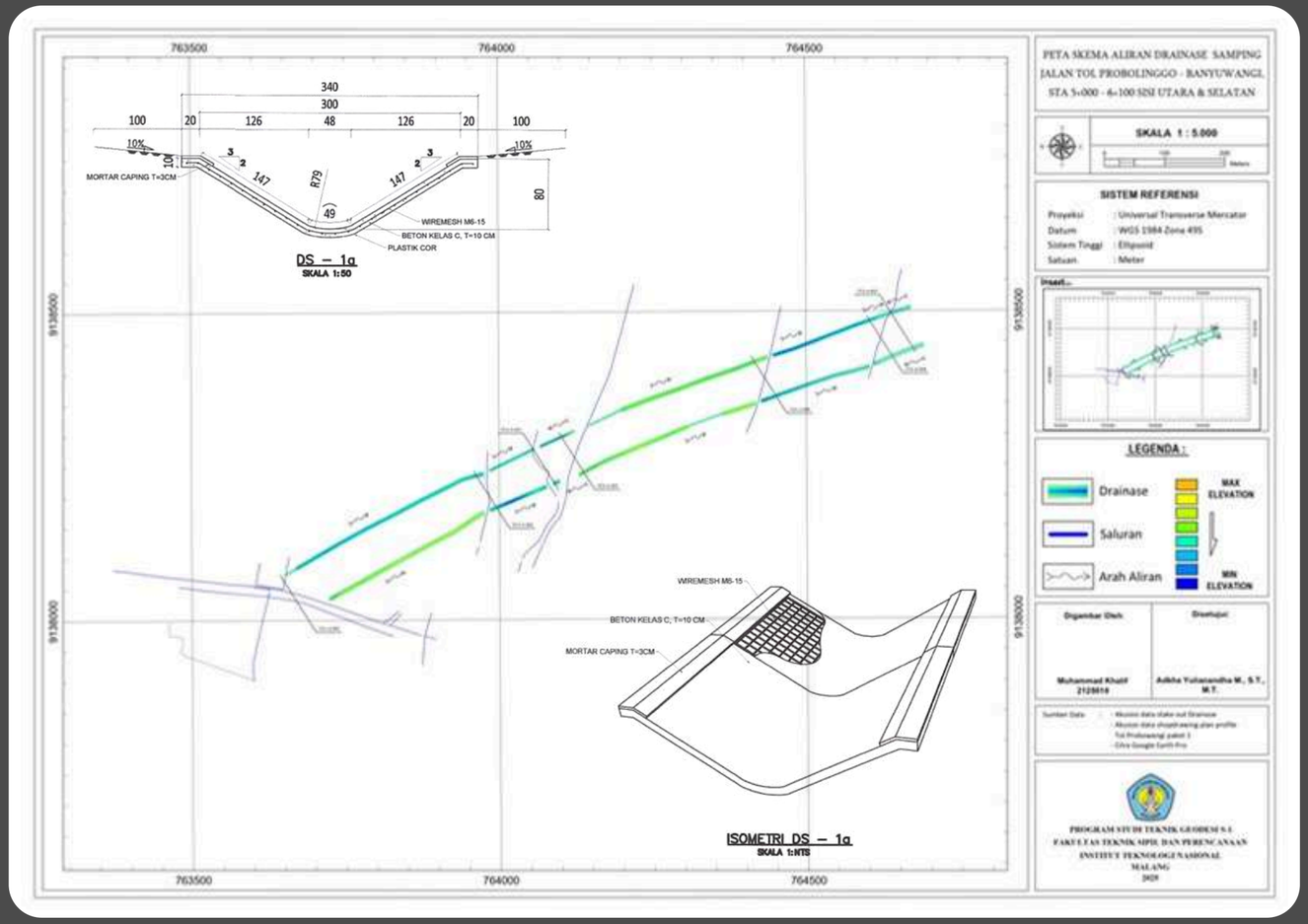
Cut & Fill Project

Station	Cut Area (Sq.m.)	Cut Volume (Cu.m.)	Reusable Volume (Cu.m.)	Fill Area (Sq.m.)	Fill Volume (Cu.m.)	Cum. Cut Vol. (Cu.m.)	Cum. Reusable Vol. (Cu.m.)	Cum. Fill Vol. (Cu.m.)	Cum. Net Vol. (Cu.m.)
0+000.066	1.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0+025.185	5.31	86.54	86.54	0.00	0.00	86.54	86.54	0.00	86.54
0+050.459	5.74	139.75	139.75	0.00	0.00	226.29	226.29	0.00	226.29
0+075.521	6.30	150.93	150.93	0.00	0.00	377.22	377.22	0.00	377.22
0+100.979	6.03	156.95	156.95	0.00	0.00	534.17	534.17	0.00	534.17
0+126.201	5.68	147.67	147.67	0.00	0.00	681.84	681.84	0.00	681.84
0+151.428	5.81	144.97	144.97	0.00	0.00	826.81	826.81	0.00	826.81
0+176.657	5.47	142.40	142.40	0.00	0.00	969.21	969.21	0.00	969.21
0+201.874	5.34	136.32	136.32	0.00	0.00	1105.53	1105.53	0.00	1105.53
0+227.099	5.29	134.00	134.00	0.00	0.00	1239.53	1239.53	0.00	1239.53
0+252.103	5.13	130.27	130.27	0.00	0.00	1369.80	1369.80	0.00	1369.80
0+277.549	5.13	130.58	130.58	0.00	0.00	1500.38	1500.38	0.00	1500.38
0+302.767	4.88	126.20	126.20	0.00	0.00	1626.58	1626.58	0.00	1626.58





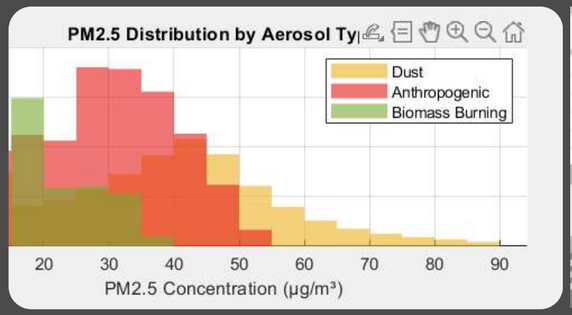
Material(s) at Station 0+060.17			
Material Name	Area	Volume	Cumulative Volume
Ground Removed	0.31	45.93	200.63
Ground Fill	0.36	3.55	3.55



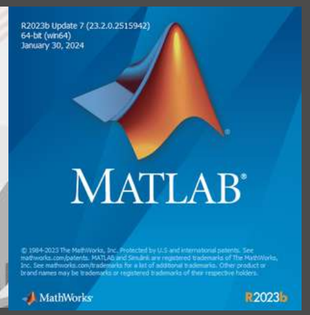
Terrestrial Survey Strategy for Determining Cut and Fill Volumes of Side Drainage in the Planning

This study aims to design a terrestrial survey strategy to support the calculation of cut and fill volumes for side drainage channels in the Probowangi Toll Road construction project, specifically at STA 5+000 – 6+000. The approach utilizes existing elevation data and planned drainage design data to analyze elevation differences between the current ground surface and the proposed drainage channel design. This calculation is essential to ensure optimal earthwork balance, improve drainage flow efficiency, and support a sustainable and cost-effective construction process.

Recent Studies

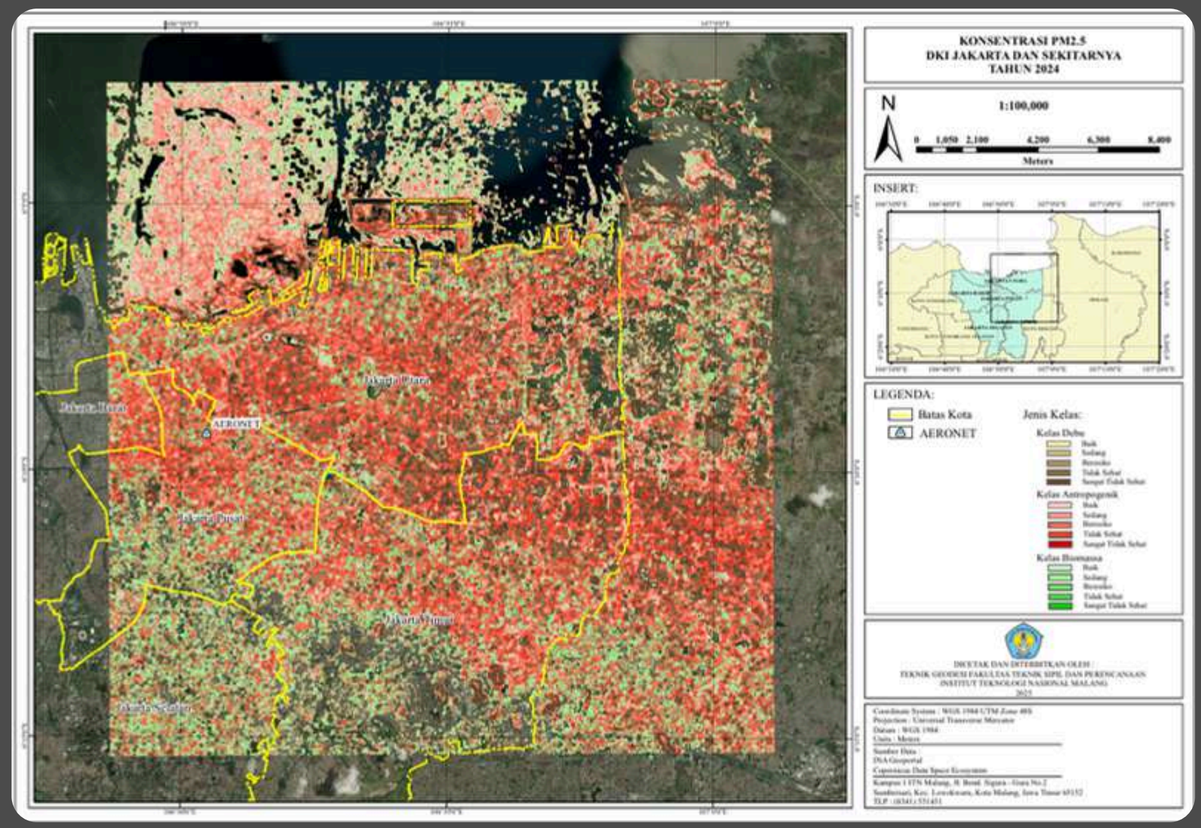






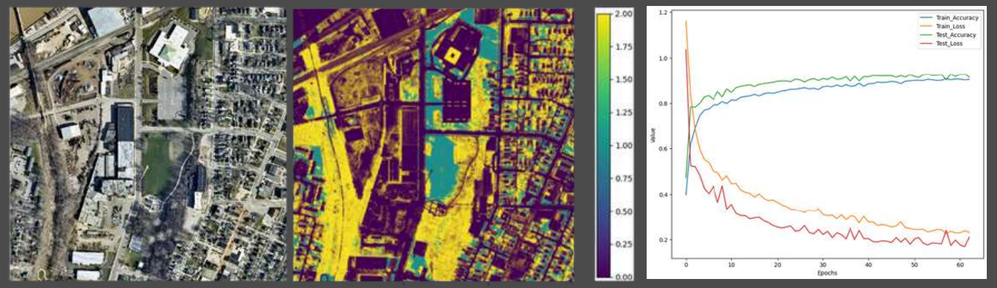
Analysis of PM2.5 Main Component Concentration Using Multitemporal Satellite Imagery and AERONET Data

This research focused on the Jakarta Metropolitan Area (DKI Jakarta and surroundings) and analyzed the concentration of PM2.5 using multi-temporal satellite data from Sentinel-2A and Landsat-8 OLI. The study applied several atmospheric analysis methods, including the Dispersion Coefficient (DC), Ångström Exponent (AE), and Normalized Gradient Aerosol Index (NGAI) to estimate aerosol distribution and identify dominant particulate components. Validation was conducted using Aerosol Optical Depth (AOD) data from the AERONET BMKG Jakarta site to ensure model accuracy and reliability.

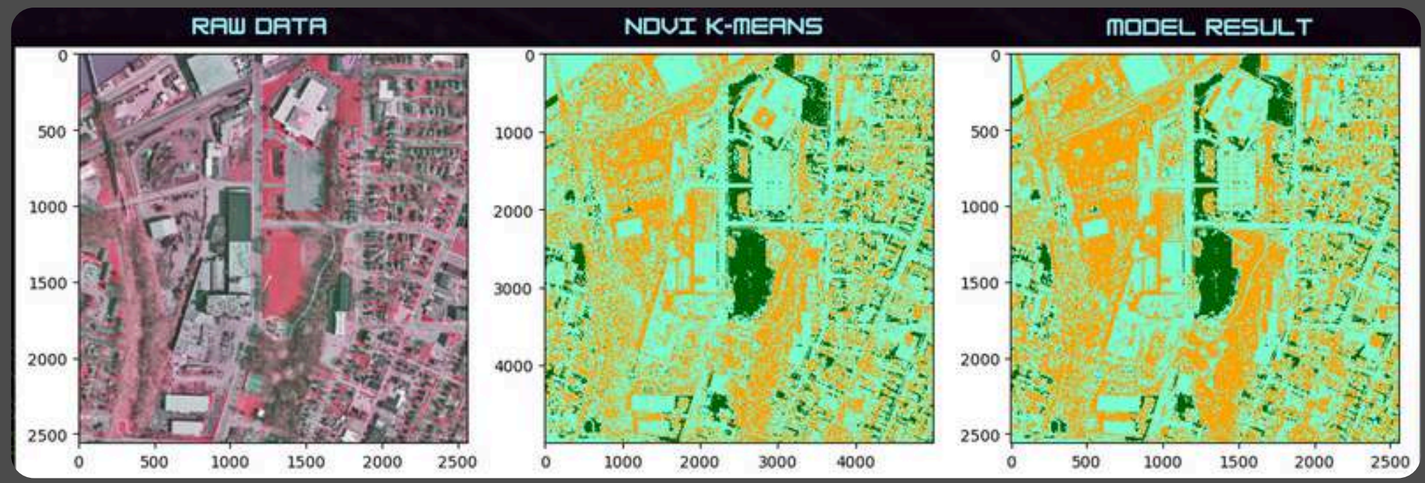


[Click here for More information about this project]

DeepLearn (AI)



	precision	recall	f1-score	support
0	0.96	0.89	0.92	938886
1	0.90	0.84	0.87	739332
2	0.82	0.95	0.88	1071984
3	0.98	0.87	0.92	395526
accuracy			0.89	3145728
macro avg	0.91	0.89	0.90	3145728
weighted avg	0.90	0.89	0.89	3145728

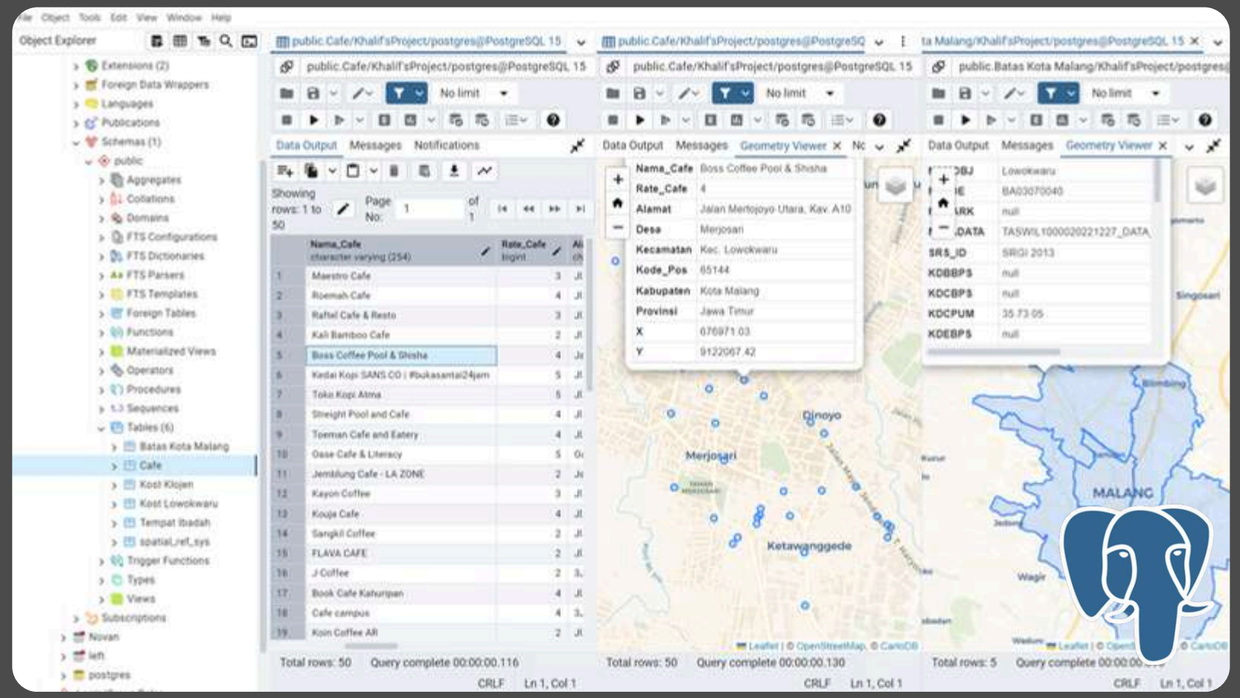


Built an NDVI-based K-Means classification model for vegetation analysis using TensorFlow and Keras.

This project focuses on leveraging remote sensing and machine learning to analyze vegetation patterns using high-resolution imagery. By applying the Convolutional Neural Networks (CNN) method, the project aims to classify and spatially map vegetation areas. The development was carried out using the Python programming language with support from libraries such as rasterio, matplotlib, numpy, pandas, scikit-image, and Pillow to facilitate raster processing, spatial visualization, and computational image feature analysis.

Click here for More information about this project

Web Map



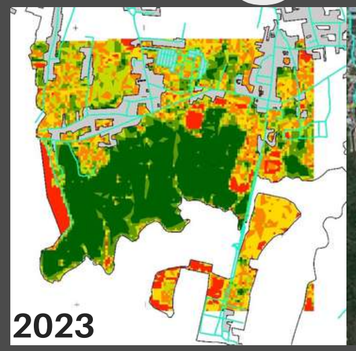


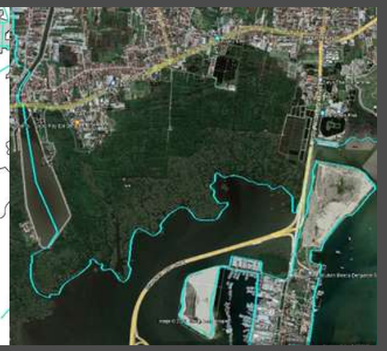
Integration of WebMap with PostgreSQL and GeoServer as the spatial database and server

This project focuses on developing an interactive web-based mapping system for managing and visualizing spatial data efficiently. The system integrates PostgreSQL as the main database, enhanced with the PostGIS extension to store and analyze geospatial data such as points, lines, and polygons. GeoServer is used to publish spatial data online through standard web services (WMS/WFS), allowing users to access, visualize, and interact with maps directly from the web without the need for specialized GIS software.

Click here for More information about this project

Mangrove Changes

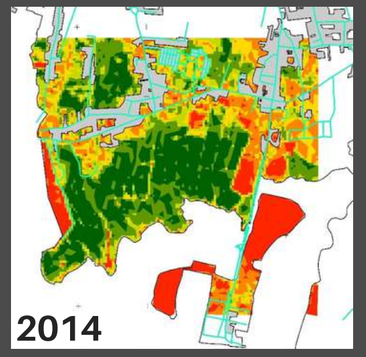


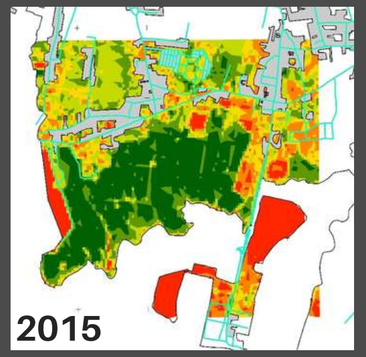


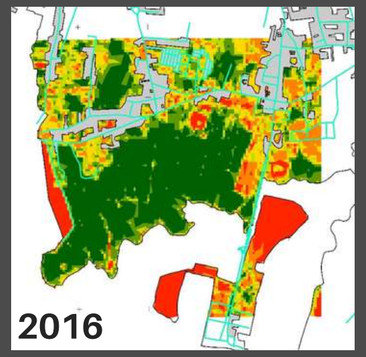


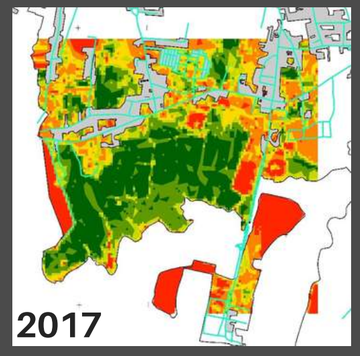
Mangrove Analysis using Google Earth Engine (Case Study: Pemogan, South Denpasar, Bali)

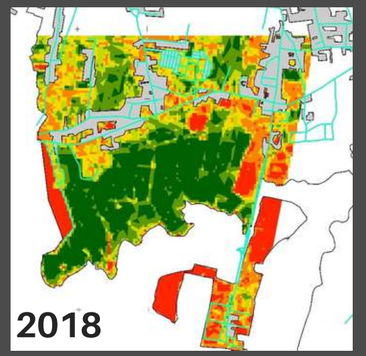
In this project, I analyzed mangrove vegetation changes from 2014 to 2023 using Google Earth Engine (GEE) and supporting GIS software. I started by collecting and processing multi-temporal satellite imagery (Landsat and Sentinel) through GEE, applying cloud masking, and calculating vegetation indices such as NDVI to measure mangrove density. The results were then exported, refined, and classified in ArcMap, including reclassification, raster-to-polygon conversion, and area calculation for each vegetation density class. To ensure accuracy, I compared the results between Landsat and Sentinel imagery and validated anomalies caused by cloud cover using alternative datasets from USGS Earth Explorer.

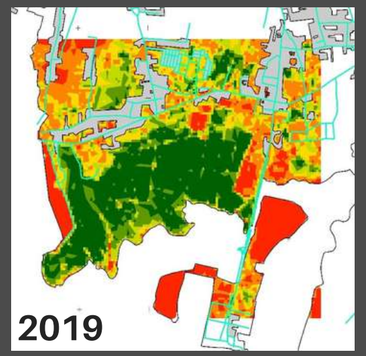


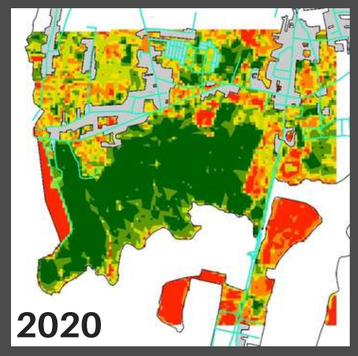


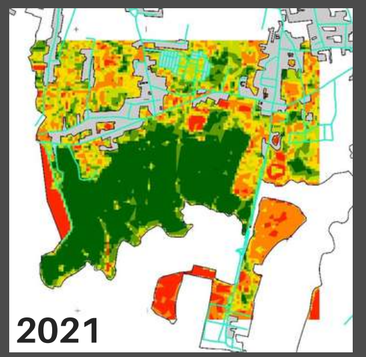


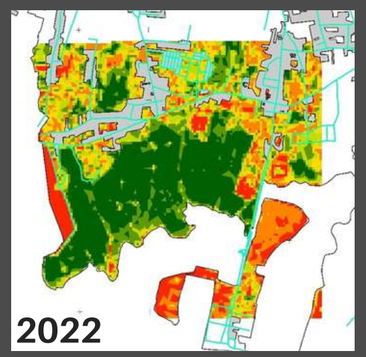






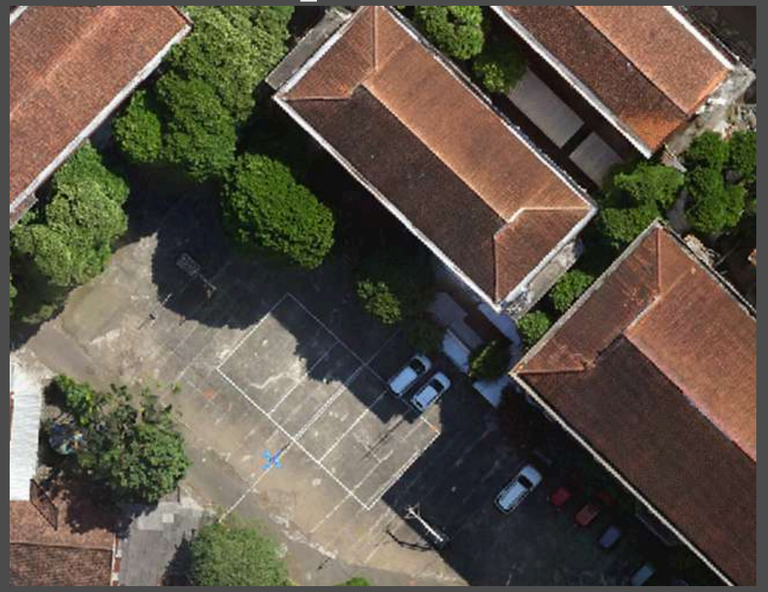






Click here for More information about this project

Orthophoto



AERIAL MAP
OF THE SUMBERSARI VILLAGE AREA
LOWOKWARU DISTRICT, MALANG CITY,
EAST JAVA, INDONESIA

Count	X error (m)	Y error (m)	Z error (m)	XY error (m)	Total (m)
4	0.113385	0.0392823	0.014451	0.119997	0.120864

Table 3. Control points RMSE.
X - Longitude, Y - Latitude, Z - Altitude.

Count	X error (m)	Y error (m)	Z error (m)	XY error (m)	Total (m)
5	0.208833	0.277197	0.938687	0.347059	1.00079

Table 4. Check points RMSE.

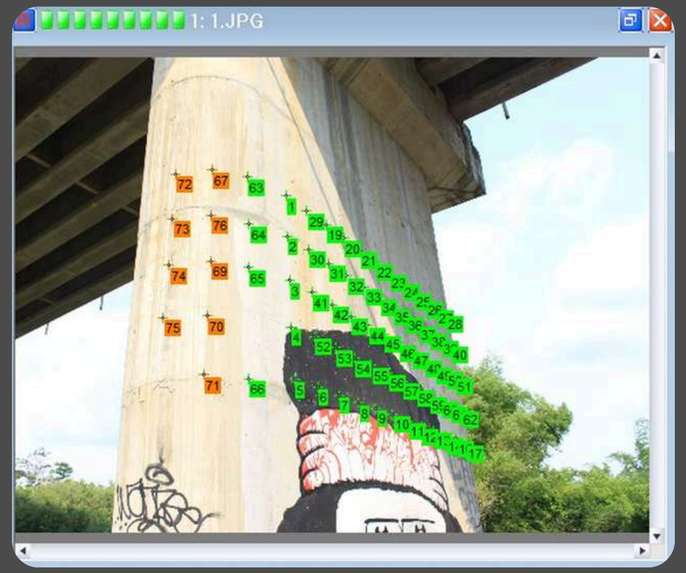


Aerial Mapping

This project was carried out by a group in which I was responsible for data processing. It focuses on orthophoto generation using aerial imagery acquired from UAVs. The workflow involved the placement of Ground Control Points (GCPs) and Independent Control Points (ICPs) to ensure spatial accuracy. The process included photo alignment, dense cloud generation, mesh construction, and the creation of Digital Elevation Models (DEM) and orthomosaics for large-scale mapping. The final outputs achieved a ground resolution of 2.09 cm/pixel, with RMSE accuracy below 1 meter for both GCP and ICP validation.

Click here for More information about this project

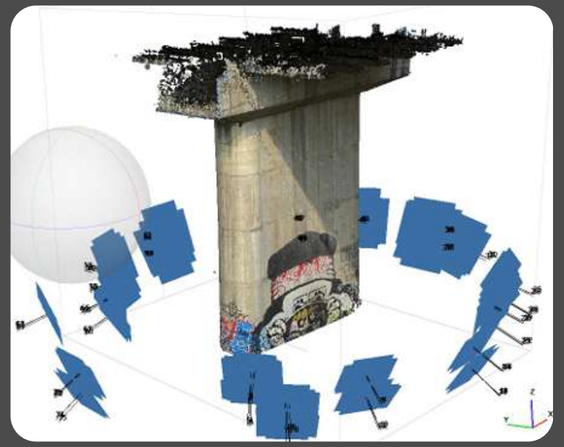
3D Modelling













3D Model

This 3D modeling project utilized a smartphone camera with convergent image acquisition and was processed using Agisoft Metashape. The main purpose of this project is to monitor and analyze structural deformation over time, focusing on detecting potential damage, measuring positional or shape changes, and ensuring the safety and integrity of infrastructures such as bridges and monuments.

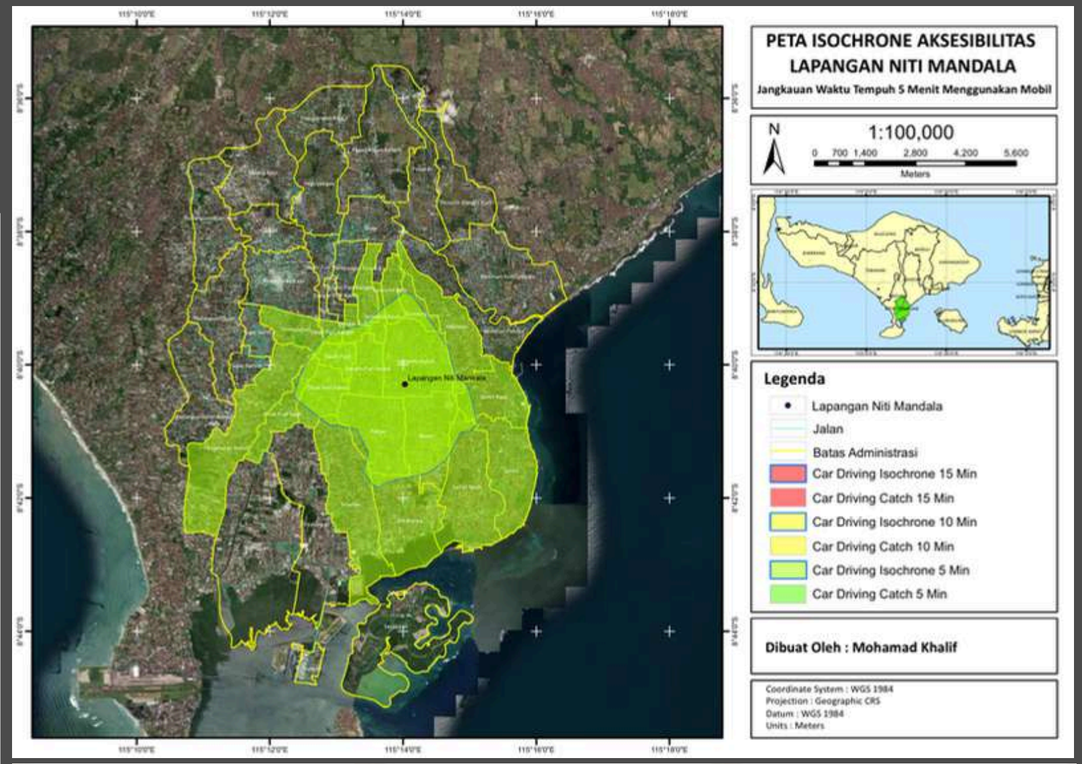
Click here for More information about this project

Isochrone-Based Accessibility Analysis

Accessibility analysis of Niti Mandala Field in Denpasar, utilizing car travel time isochrones through QGIS, ORS Tools, and Python.

This project evaluates the extent and distribution of access to a centralized public facility for residents across 43 sub-districts in Denpasar. Isochrones were selected due to their reliance on travel time within the road network, rendering them more realistic than buffers or isodistances. The analysis was conducted via the Python console in QGIS utilizing the ORS Tools plugin (openrouteservice) for travel times of 5, 10, and 15 minutes, subsequently overlaid with sub-district boundaries and population data. The population within the catchment area was determined using the formula POP_CATCH = population × catchment area fraction.

Primary findings:
Reached population: 23.1% (within 5 minutes), 74.6% (within 10 minutes), and 97.3% (within 15 minutes).
The coverage pattern creates concentric rings surrounding the field. The initial ring (<5 minutes) encompasses Dauh Puri, Panjer, Sumerta Kelod, and Dangin Puri Kelod. The third ring includes the northern and southwestern peripheries, which achieve complete coverage only after 15 minutes.
The area covered expanded from 15.8 km² (5 minutes) to 76.5 km² (10 minutes). The coverage extends in a north-south direction along the road corridor.
The most challenging villages to access are Peguyangan Kaja, Serangan, and Sanur in the southeast. Serangan and Sanur face obstacles due to waterways and narrow paths, highlighting the need for route validation in these areas.











Click here for More information about this project

Certificate of Internship



SERTIFIKAT KERJA PRAKTIK

Diberikan Kepada :

Mohamad Khalif

MAHASISWA PROGRAM STUDI TEKNIK GEODESI S-1
FAKULTAS TEKNIK SIPIL DAN PERENCANAAN
INSTITUT TEKNOLOGI NASIONAL MALANG
ATAS PARTISIPASI DALAM PEMBANGUNAN JALAN TOL
PROBOLINGGO - BANYUWANGI PAKET 1

Dalam kegiatan Kerja Praktik 2024 yang diselenggarakan pada
25 Juli 2024 - 25 September 2024
oleh PT. VIRAMA KARYA

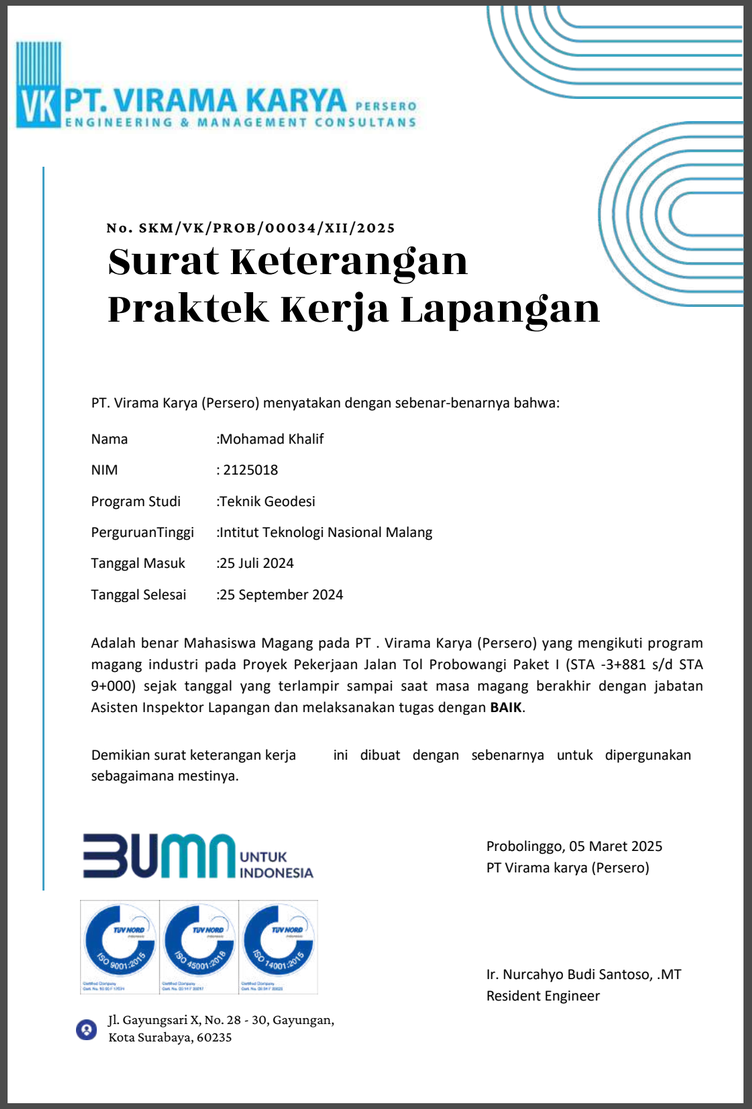
Mengetahui,

Ketua Program Studi
Teknik Geodesi

Ir. Dedy Kurnia Sunaryo, M.T.
NIP.Y.10330600413

RESIDENT ENGINEER

Ir. Nurcahyo Budi Santoso, M.T.



PT. VIRAMA KARYA PERSERO
ENGINEERING & MANAGEMENT CONSULTANS

No. SKM/VK/PROB/00034/XII/2025

Surat Keterangan Praktek Kerja Lapangan

PT. Virama Karya (Persero) menyatakan dengan sebenar-benarnya bahwa:

Nama	:Mohamad Khalif
NIM	: 2125018
Program Studi	:Teknik Geodesi
PerguruanTinggi	:Intitut Teknologi Nasional Malang
Tanggal Masuk	:25 Juli 2024
Tanggal Selesai	:25 September 2024

Adalah benar Mahasiswa Magang pada PT . Virama Karya (Persero) yang mengikuti program magang industri pada Proyek Pekerjaan Jalan Tol Probowangi Paket I (STA -3+881 s/d STA 9+000) sejak tanggal yang terlampir sampai saat masa magang berakhir dengan jabatan Asisten Inspektor Lapangan dan melaksanakan tugas dengan **BAIK**.

Demikian surat keterangan kerja ini dibuat dengan sebenarnya untuk dipergunakan sebagaimana mestinya.

Probolinggo, 05 Maret 2025
PT Virama karya (Persero)

Ir. Nurcahyo Budi Santoso, .MT
Resident Engineer

Jl. Gayungsari X, No. 28 - 30, Gayungan,
Kota Surabaya, 60235

Certificate of Participation


SERTIFIKAT
041/B.3/HMG-I/XI/2023
Di Berikan Kepada :
Mohamad Khalif
Atas Partisipasinya Sebagai :
Anggota Publikasi Dekorasi Dokumentasi
Kegiatan Dies Natalis Teknik Geodesi ITN Malang Ke-38 Tahun 2023 dengan tema "Geodesy For Indonesian Sustainable Development" Yang Diselenggarakan Pada Tanggal Rabu, 26-27 Juli 2023
Ketua Program Studi Teknik Geodesi ITN Malang
Silvester Sari Sai, S.T., M.T.
NIP.Y. 1030600413
Ketua Himpunan Mahasiswa Geodesi
Jusman Prahara
NIM. 20.250.55
Ketua Pelaksana Diesnatalis Teknik Geodesi Ke-38
Radian Anugrah
NIM. 20.25.034


SERTIFIKAT
No. 100/RDS/IMGI/XII/2022
MENGUCAPKAN TERIMAKASIH KEPADA
Muhammad Khalif
Sebagai Koordinator Sie Dokumentasi & Dekorasi penyelenggara program kerja IMGI
DIES NATALIS IMGI XXIX
Yang diselenggarakan pada tanggal
3 Desember 2022
Candra Dwi Wibowo
Sekretaris Jenderal IMGI


HIMPUNAN MAHASISWA GEODESI
FAKULTAS TEKNIK SIPIL DAN PERENCANAAN
INSTITUT TEKNOLOGI NASIONAL MALANG
SERTIFIKAT
003/B.3/HMG-I/XI/2021
Diberikan Kepada:
Mohamad Khalif
2125018
Atas Partisipasinya Sebagai :
PESERTA
Dalam "Geodesi Menyapa (GEMA)" Dengan Tema
From Virtual To Mutual
"Menciptakan Lingkungan Kekeluargaan Yang Membantu Pembentukan Karakter Geodet Muda"
Yang Diselenggarakan Pada 04 Desember 2021 - 19 Maret 2022
Melalui Room Zoom Meeting
Kaprodi Teknik Geodesi ITN Malang
Silvester Sari Sai, S.T., M.T.
NIP.Y.1030600413
Ketua Himpunan Mahasiswa Geodesi
Nurkhamid Hidayat
NIM. 19.25.012
Ketua Pelaksana
Dawud Wahyuda N.Daya
NIM. 19.25.004


SERTIFIKAT
004/B.3/HMG-I/X/2022
Di berikan Kepada :
Muhammad Khalif
Atas Partisipasinya Sebagai :
PANITIA
Dalam Geodesi Menyapa (GEMA) yang bertemakan
"Memperkenalkan dan mewujudkan mahasiswa yang profesional dalam solidaritas serta kreativitas pada era milenial"
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ATAS PARTISIPASINYA SEBAGAI:
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Dalam Kegiatan *Study Excursion* 2023 dengan tema
"Mewujudkan Generasi Muda Yang Berkompeten Dan Siap Menjadi Surveyor Sukses"
yang diselenggarakan pada tanggal, 21-24 Februari 2023 oleh HMG ITN MALANG
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TOEFL Prediction



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LANGUAGE LABORATORY UNIT
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CERTIFICATE

OF ACHIEVEMENT

NO. 1578/TPT/lab.bhs/ITN/VIII/2025

This certifies that

Name : MOHAMAD KHALIF

Registration Number: 2125018

Test Date : 11 August 2025

has taken a PREDICTION TEST for the TOEFL administered by the Language Laboratory Unit - ITN Malang.

SCALED SCORES

Listening Comprehension	48
Structure and Written Expression	42
Reading Comprehension	48
Overall Score	460

Head,

Rini Anjarwati, S.Pd.,M.Pd

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